

Representation of Symmetric groups

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Contents

1	Some facts on linear representation theories	2
2	Irreducible representations on S_n	2
2.1	Reviewing facts on S_n	2
2.2	Illustrations	3
2.2.1	Permutation of Young tableaux	4
2.2.2	Dimensions of representations from Young tableaux	5
3	Representations of S_n and other groups	5
4	Schur-Weyl	5

1 Some facts on linear representation theories

2 Irreducible representations on S_n

2.1 Reviewing facts on S_n

In S_n , every element $\sigma \in S_n$ is called a permutation. The first fact on the decomposition of σ is that σ can be decomposed into some disjoint cycles, called **cycle decomposition** of σ . This decomposition uniquely exists guaranteed by the following:

Corollary 2.0.1 (Existence and uniqueness of the cycle decomposition) *Every $\sigma \in S_n$ can be uniquely expressed as a cycle decomposition, up to the order of factors.*

This existence and uniqueness permits us to give the following definition

Definition 2.1: Cycle type

For every element $\sigma \in S_n$, σ has a unique decomposition

$$\sigma = \tau_1 \tau_2 \cdots \tau_r$$

into disjoint cycles $\tau_1, \tau_2, \dots, \tau_r$ arranged in decreasing order of cycle length. Let k_i be the length of τ_i . The r -tuple $(k_1, k_2, \dots, k_r)$ is called the **cycle type** of σ .

For S_n , the (conjugate) action of S_n onto itself gives the orbit of S_n whose elements are those conjugacy classes of S_n . These conjugacy classes of S_n are synonyms as these cycle types.

Theorem 2.1: conjugacy classes and cycle types

The conjugacy classes of S_n is bijectively correspondent to the cycle types of S_n .

Definition 2.2: Partition of integers

Let $n \in \mathbb{Z}_+$ and $1 \leq r \leq n$. A sequence $\lambda = (\lambda_1, \lambda_2, \dots, \lambda_r)$ is a **partition** of n , denoted $\lambda \vdash n \Leftrightarrow \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_r \geq 1$ and $\sum_i \lambda_i = n$.

In this case,

- λ_i s are the *parts* of the partition.
- r is the *length* of the partition, denoted $l(\lambda)$.
- n is the *size* of the partition, denoted $|\lambda|$.

The set of all partitions of n , $\{\lambda | \lambda \vdash n\}$ is denoted by $\wp(n)$, namely $\wp(n) = \{\lambda | \lambda \vdash n\}$.

Remark This notion can be extended to 0 by conventions, $\wp(0) = \{\emptyset\}$ because the sequences are required to be nonzero. So, no sequences represent 0.

Example *Cycle types of elements of S_n are partitions of n .*

Definition 2.3: Conjugate partitions

Let $\lambda = (\lambda_1, \lambda_2, \dots, \lambda_k) \vdash n$ be a partition of $n \in \mathbb{Z}_+$. The **conjugate partition** of λ , denoted λ' , is defined to be $\lambda' = (\mu_1, \mu_2, \dots, \mu_{\lambda_1})$ where $\mu_j = |\{i \in [k] : j \leq \lambda_i\}|$.

Remark The intuition of the conjugate is illustrated in the following example using Young diagram 2.5. After the rotation of $Y_{(4)}$, it coincides with $Y_{(1,1,1,1)}$. Similar cases happen to $Y_{(3,1)}$ and $Y_{(2,1,1)}$. Conjugation is a formalization of each pair of relation.

What kind of relation is satisfied in each pair?

- The length of λ' , $l(\lambda')$ should be the maximum of $\{j | (i, j) \in Y_\lambda\}$, which is λ_1 .
- Each column of $Y_{\lambda'}$ should be a row in Y_λ . Hence, μ_j is the cardinality of the points $\{(i, j) \in Y_\lambda | 1 \leq i \leq l(\lambda) = k, j \leq \lambda_i\}$ for a fixed j , which is $|\{i \in [k] : j \leq \lambda_i\}|$.

How to get the conjugate partition $\lambda' \vdash n$ out of the partition $\lambda \vdash n$?

Definition 2.4: Dominant relation

Let $\lambda = (\lambda_1, \dots, \lambda_k) \vdash n$ and $\mu = (\mu_1, \dots, \mu_s) \vdash n$ be partitions of n .
 λ **dominates** μ , denoted $\mu \preceq \lambda$ or $\lambda \succeq \mu \Leftrightarrow \forall l \in [\min(k, s)], \sum_{i=1}^l \lambda_i \geq \sum_{i=1}^l \mu_i$.

Remark The intuition of dominance for λ and μ in the Young diagram languages is the comparison of the number of blocks in each row within the least row numbers between Y_λ and Y_μ .

Example When $n = 4$, $(4) \succeq (3, 1) \succeq (2, 2) \succeq (2, 1^2) \succeq (1^4)$.

Corollary 2.1: Dominance on $\wp(n)$

he dominance is a partial ordering on $\wp(n)$. It extends to the lexicographic ordering (total ordering).

Proof:

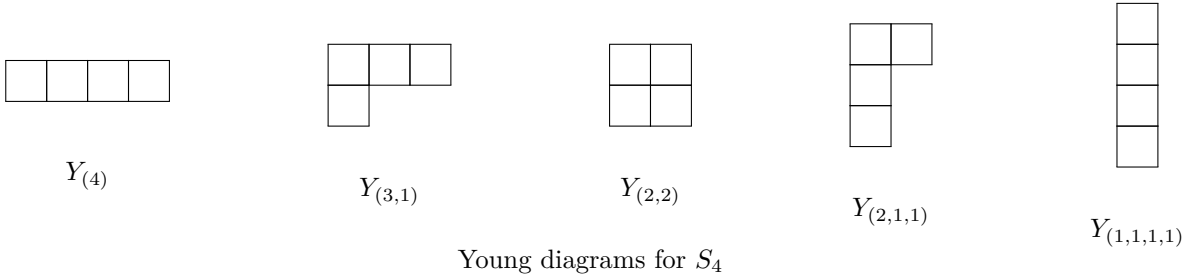
□

2.2 Illustrations**Definition 2.5: Young diagram**

Let $\lambda = (\lambda_1, \lambda_2, \dots, \lambda_r) \vdash n$ be a partition of integer n . The **Young diagram** of λ , denoted Y_λ , is defined to be the following set

$$Y_\lambda = \{(i, j) \in \mathbb{Z}_+ \times \mathbb{Z}_+ | 1 \leq i \leq r, 1 \leq j \leq \lambda_i\}$$

Example Young diagrams has an illustration version. Take S_4 for example. Corresponding to the cycle types of S_4 : (4) , $(3, 1)$, $(2, 2)$, $(2, 1, 1)$ and $(1, 1, 1, 1)$, the followings are the well-known Young diagrams for S_4 .



So, Young diagrams provide a realization of partition of certain integers. Now, the following relations are established:

irreducible representations on $S_n \rightsquigarrow$ Cycle types of $S_n \rightsquigarrow$ partition of integers $\rightsquigarrow$ Young diagrams

Definition 2.6: Young tableau

Still, let $\lambda \vdash n$ be a partition of n and Y_λ be a Young diagram set as definition 2.5. Then, the λ -**tableau** or **Young tableau** of the shape λ (with respect to the partition $\lambda \vdash n$ or with respect to the Young diagram Y_λ) is a bijection

$$t : Y_\lambda \rightarrow [n]$$

where $[n] := \{1, 2, \dots, n\}$.

The set of all λ -tableau is denoted by Δ^λ .

Example Illustrations of Young tableaux.

Why do we introduce the Young tableaux?

Definition 2.7: Standard and normal Young tableau

A **standard Young tableau** with respect to $\lambda \vdash n$ is a function t satisfying the following two conditions:

- (i) (row increasing): for any (i, j) and $(i, j + 1)$ in Y_λ , $t(i, j) < t(i, j + 1)$.
- (ii) (column increasing): for any (i, j) and $(i + 1, j)$ in Y_λ , $t(i, j) < t(i + 1, j)$.

A **canonical Young tableau** with respect to $\lambda \vdash n$, denoted t^λ , is *the* strictly monotonically increasing function when Y_λ is equipped with the lexicographic order.

Remark The article *the* appear in the definition of canonical Young tableau is because the following corollary in set theory:

Strict order-preserving bijections between two finite totally ordered sets with equal cardinality uniquely exist.

When Y_λ is assigned a lexicographic order, Y_λ is a totally ordered set.

Example (1) A normal Yong tableau must be a standard Young tableau.

(2) For S_4 , a standard Young tableau might be:

1	2	4
3		

1	3	4
2		

While, the mere normal Young tableau of $(3, 1)$ is:

1	2	3
4		

2.2.1 Permutation of Young tableaux

Let's consider the actions of S_n on Young tableaux. The transitive action $S_n \curvearrowright [n]$ induces an action $S_n \curvearrowright \Delta^\lambda$ as follows. For any $\sigma \in S_n$ and any Young tableau $t : Y_\lambda \rightarrow [n]$ with respect to some λ , the action ${}^\sigma t$ is elementwise defined to be

$$({}^\sigma t)(i, j) := \sigma(t(i, j))$$

The intuition for this action is: if the slot (i, j) in Y_λ has the integer $k = t(i, j)$, then the slot (i, j) has $\sigma(k)$ after the action σ on t .

Consider the collection of each rows and each column. For a fixed i ,

$$R_i(t) := \{t(i, j) \in Y_\lambda : 1 \leq j \leq \lambda_i\}$$

To describe the column elements, columns are rows in the conjugate Young diagram of Y_λ . Thereby, for a fixed j ,

$$C_j(t) := \{t(i, j) \in Y_\lambda : 1 \leq i \leq \lambda'_j\}$$

Then, this action $S_n \curvearrowright \Delta^\lambda$ induces actions on each $R_i(t)$ and $C_j(t)$. For example, ${}^\sigma R_i(t) := \{({}^\sigma t)(i, j) : 1 \leq j \leq \lambda_i\}$. If two $s, t \in \Delta^\lambda$ making each row $R_i(t)$ stable, this could be equivalent in some sense. More precisely,

$$s \sim_R t \Leftrightarrow \forall i, R_i(s) = R_i(t)$$

This is indeed an equivalence relation. The corresponding equivalence class of t is denoted $[t]_R$, and the quotient space is denoted $\Omega_R^\lambda = \Delta^\lambda / \sim_R$, called the λ -tabloid of rows.

For each $[t]_R \in \Omega_R^\lambda$, its stabilizer is

$$\begin{aligned} \text{Stab}_{S_n}([t]_R) &= \{\sigma \in S_n : {}^\sigma [t]_R = [t]\} \\ &= \{\sigma \in S_n : {}^\sigma t \sim_R t\} \\ &= \{\sigma \in S_n : \forall i, R_i({}^\sigma t) = R_i(t)\} \end{aligned}$$

On the other hand, we define the *row stabilizer* of t , denoted $R(t)$,

$$R(t) = \{\sigma \in S_n : \forall i, {}^\sigma R_i(t) = R_i(t)\}$$

in which we would like all elements $\sigma \in S_n$ that fixes each row elements. Via the equivalence ${}^\sigma R_i(t) = R_i({}^\sigma t)$ for each i , the stabilizers of $[t]_R$ is the same as the row stabilizer of t :

$$R(t) = \text{Stab}_{S_n}([t]_R)$$

Similarly, this setting has a duplication for columns. One can define $\sim_C$, $[t]_C$, $C(t)$, Ω_C^λ and $C(t) = \text{Stab}_{S_n}([t]_C)$.

Since we just know ${}^\sigma R(t) = R({}^\sigma t)$, what is the relation between ${}^\sigma R(t)$ and $R(t)$? The answer is like the scenario among different orbits, ${}^\sigma R(t) = \sigma R(t) \sigma^{-1}$.

Definition 2.8: Young symmetrizer

The **row symmetrizer** is

$$a_\lambda := \sum_{\sigma \in R(t)} \sigma$$

Remark This element is in the group ring $\mathbb{C}[S_n]$. It is like the 'averaging' of elements of S_n that makes every row stable.

Example Consider the case of $n = 3$ and $\lambda = (2, 1)$. The standard Young tableau of $(2, 1)$, $t^{(2,1)}$ is

$$t^{(2,1)} = \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 & \\ \hline \end{array}$$

The vertical permutations are $(1), (13)$ and the horizontal permutations are $(1), (12)$.

2.2.2 Dimensions of representations from Young tableaux

Is it possible to read the dimension of irreducible representations from a Young tableau $t : Y_\lambda \rightarrow [n]$? If possible, how to read that? The answer is the hook length formula.

3 Representations of S_n and other groups

4 Schur-Weyl